Small-time Controllability of Control Systems on 3D Lie Groups

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Contents

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Project title: Small-time controllability of control systems on 3D Lie groups

Project description: The goal of the project is to determine small-time controllability of affine control systems with one control on 3D unimodular Lie groups. By a parameter count one can check that all such control systems are equivalent to a finite number of model ones. After the classification is done, minimum controllability time will be estimated. This problem has some applications in quantum control, where the minimum controllability time is called quantum speed limit.

1 Introduction

As motivation, the Schrodinger equation is

$$i\dot{\psi} = -\frac{\Delta}{2}\psi + \sum u_i(t)V_i(t)$$

In this project, we are studying the control system

$$\dot{g} = (A + uB)g, \quad G \subset GL(n) \tag{1.1}$$

where u is the control input. We are particularly interested in working with the case $dimG = 3$, the unimodular Lie groups of dimension 3, which can be completely classified into five types of Lie groups: $SL(2)$, $SU(2)$, H^3, $SE(2)$, and $SH(2)$.

This project involves two key goals

1. Classify systems of the type 1.1 on all unimodular Lie groups of dimension 3
2. Find all examples of this system 1.1 which are controllable in arbitrary small time

Todo: need to give a lot more motivation.

2 Preliminaries

2.1 Groups

We first give the basic notions of groups, along with some examples.

Definition 2.1. *A group G is a non-empty set with an operation $G \times G \to G$, $(a, b) \mapsto ab$ such that the following axioms hold:*

1. $(ab)c = a(bc)$ *(associativity)*
2. $\exists e \in G$ *such that* $eg = ge$ *for all* $g \in G$ *(identity)*
3. $\forall g \in G$, $\exists d \in G$ *such that* $dg = gd = e$ *(inverse)*

A group is called commutative or Abelian if for all $g, h \in G$, we have $gh = hg$.

Example 2.2. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{F}$, $M_2(\mathbb{F})$, $\mathbb{Z}/n\mathbb{Z}$, *all with the addition operation, are groups.*

Example 2.3. *Some more examples of groups*

- *Permutation group* S_n
- *Dihedral group* D_8
- *General linear group* $GL(n, \mathbb{F})$

Definition 2.4. *Let G be a group. A subset $H \subset G$ is called a subgroup if*

1. $e \in H$
2. $a, b \in H \Rightarrow ab \in H$
3. $a \in H \Rightarrow a^{-1} \in H$

A subgroup H is a group in its own right.

Definition 2.5. *Let G be a group. G is called cyclic if there is a $g \in G$ such that any element of G is a power of g, i.e. $G = \{g^n : n \in \mathbb{Z}\}$. This element g is called a generator of G.*

Next, we introduce group actions on sets.

Definition 2.6. *Let G be a group and let S be a non-empty set. G acts on S if we are given a function $G \times S \to S$, $(g, s) \mapsto gs$ such that*

1. $es = s$, $\forall s \in S$
2. $(g_1 g_2)s = g_1(g_2 s)$, $\forall g_1, g_2 \in G, s \in S$

Definition 2.7. *We are given an action of G on S, and an element $s \in S$ The orbit of s is*

$$Orb(s) = \{gs : g \in G\}$$

The stabiliser of s is

$$Stab(s) = \{g \in G : gs = s\}$$

Note that $Orb(s)$ is a subset of S, while $Stab(s)$ is a subset (in fact a subgroup) of G. Below are some examples.

Example 2.8. *Let the group be $G = S_n$, let the set be $T = \{1, 2, \ldots, n\}$. Then for $i \in T$, $\sigma i = \sigma(i)$, and $Orb(i) = T$.*

Example 2.9. *Let the group be $G = \mathbb{R}$, let the set be $S =$ sphere in $\mathbb{R}^3$ of radius 1 about the origin. $\mathbb{R}$ acts by rotating the sphere around the z-axis. The element $r \in \mathbb{R}$ rotates r radians. For $s \in S$, $Orb(s)$ is the altitude line on the sphere.*

Example 2.10. *Let G be a group, let $H \subset H$ be a subgroup. Here, we let H be the group and G be the set. H acts on G by $H \times G \to G$, $(h, g) \mapsto hg$. For $g \in G$, $Orb(g) = \{hg : h \in H\} = Hg =$ right cosets of H.*

Definition 2.11. *A group action of G on X is called transitive if $\forall x, y \in S$, $\exists g \in G$ such that $gx = y$.*

2.2 The Exponential Map

Let $M_n(\mathbb{F})$ be the space of $n \times n$ matrices with coefficients in the field $\mathbb{F}$ ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$). We consider it as a normed vector space with norm

$$||X|| = \sup_{u \neq 0} \frac{|X(u)|}{|u|}$$

where $|u|$ is the usual norm in $\mathbb{R}^n$ or the hermitian norm in $\mathbb{C}^n$.

Definition 2.12. *For $X \in M_n(\mathbb{F})$, we define*

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

Using the Cauchy-Schwartz inequality and the property $||XY|| \leq ||X||||Y||$, we can conclude that this series is normally convergent on compact sets.

Theorem 2.13. *Let $X \in M_n(\mathbb{R})$ be a fixed matrix. Then the map $\exp : \mathbb{R} \to M_n(\mathbb{R})$ defined by*

$$t \to \exp(tX)$$

is analytic and is the unique solution to the differential equation

$$\frac{dY}{dt} = XY$$

with initial condition $Y(0) = Id$.

Proposition 2.14. *The exponential map satisfies the following properties*

1. *For any $g \in GL(n, \mathbb{R})$ and $X \in M_n(\mathbb{R})$,*

$$g(\exp(X))g^{-1} = \exp(gXg^{-1})$$

2. *If $X, Y \in M_n(\mathbb{R})$ are two commuting matrices then*

$$\exp(X + Y) = \exp(X)\exp(Y)$$

3. *For any $X \in M_n(\mathbb{R})$*

$$\exp(-X) = (\exp(X))^{-1}$$

Proposition 2.15 (Jacobi's formula)**.** *For any $X \in M_n(\mathbb{F})$ we have*

$$\det(\exp X) = e^{tr(X)}$$

2.3 Lie Groups, Lie Algebras

Definition 2.16. *A Lie group is a topological group which is a smooth manifold and such that the group operations, product $G \times G \to G$ and inverse map $G \to G$*

$$(g, h) \mapsto gh \text{ and } g \mapsto g^{-1}$$

are smooth.

Proposition 2.17. *The group of invertible matrices of rank n, $GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\}$, is a Lie group.*

We will mostly be dealing with Lie groups which are submanifolds of $GL(n, \mathbb{R})$, which are called linear Lie groups.

Definition 2.18. *Let G and H be Lie groups. A Lie group morphism from G to H is a smooth map $f : G \to H$ which is a group homomorphism.*

A Lie group isomorphism between G and H is a smooth diffeomorphism $f : G \to H$ which is a group isomorphism. When $G = H$, we call f a Lie group automorphism.

Definition 2.19. *A linear representation of G is a Lie group morphism $f : G \to GL(V)$ where V is a finite dimensional real vector space.*

A matrix representation of G is a Lie group morphism $f : G \to GL(n, \mathbb{R})$ or $f : G \to GL(n, \mathbb{C})$ for some $n \in \mathbb{N}$.

A representation of G is either a linear representation or a matrix representation.

Next, we move on to the topic of Lie algebras.

Definition 2.20. *A Lie algebra over a field $\mathbb{F}$ is a vector space $\mathfrak{g}$ over $\mathbb{F}$ equipped with a bilinear map (called a Lie bracket)*

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

such that

1. $[X, Y] = -[Y, X]$ *(skew-symmetry)*
2. $[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0$ *(Jacobi identity)*

Definition 2.21. *A Lie subalgebra of $\mathfrak{g}$ is a vector subspace $\mathfrak{h}$ such that for all $X, Y \in \mathfrak{h}$, $[X, Y] \in \mathfrak{h}$.*

Example 2.22. *The vector space $M_n(\mathbb{F})$ is a Lie algebra with the Lie bracket defined for all $X, Y \in M_n(\mathbb{F})$ by $[X, Y] = XY - YX$.*

To any Lie group, we can associate a Lie algebra. It is the tangent space at the identity element of the group. First, in the case of linear Lie groups, we can develop this idea without knowledge of the tangent space to a manifold.

Definition 2.23. *Let $G \subset GL(n, \mathbb{R})$ be a linear group. Define*

$$Lie(G) = \{X \in M_n(\mathbb{R}) : \exp(tX) \in G \; \forall t \in \mathbb{R}\}$$

Proposition 2.24. *Let $G \subset GL(n, \mathbb{R})$ be a linear group. Then $Lie(G) \subset M_n(\mathbb{R})$ is a Lie subalgebra.*

Definition 2.25. *Let G be a linear Lie group and let $\mathfrak{g}$ be its Lie algebra.*

1. *The adjoint representation of the Lie group is the homomorphism*

$$Ad : G \to Aut(\mathfrak{g})$$

given by $Ad(g) = Ad_g$.

2. *The adjoint representation of the Lie algebra is the homomorphism*

$$ad : \mathfrak{g} \to End(\mathfrak{g})$$

given by $ad(X) = ad_X$.

We refer to Ad_g as the adjoint map defined by g, and to Ad as the adjoint map. We say the derivation ad_X is the adjoint map defined by X. The following gives a relation between these maps.

Proposition 2.26. *Let G be a linear Lie group, let $\mathfrak{g}$ be its Lie algebra, and $X \in \mathfrak{g}$. Then*

$$\frac{d}{dt} Ad_{\exp(tX)|_{t=0}} = ad_X$$

Definition 2.27. *Let $\mathfrak{g}$, $\mathfrak{h}$ be Lie algebras over a field K. A Lie algebra morphism from $\mathfrak{g}$ to $\mathfrak{h}$ is a K-linear map $\phi : \mathfrak{g} \to \mathfrak{h}$ such that $\forall X, Y \in \mathfrak{g}$, $[\phi(X), \phi(Y)] = [X, Y]$.*

A Lie algebra isomorphism is a bijective Lie algebra morphism (its inverse is also a Lie algebra morphism). When $\mathfrak{g} = \mathfrak{h}$, a Lie algebra isomorphism is called a Lie algebra automorphism.

Definition 2.28. *A Lie subalgebra of $\mathfrak{g}$ is a vector subspace $V \subset \mathfrak{g}$ such that $[X, Y] \in V$ for all $X, Y \in V$.*

Definition 2.29. *A representation of a Lie algebra $\mathfrak{g}$ is a Lie algebra morphism $\phi : \mathfrak{g} \to End(V)$ where V is a vector space.*

Definition 2.30. *Any vector space V can be endowed with the Lie bracket defined by $[X, Y] = 0$ for all $X, Y \in V$. Such a Lie algebra $(V, [\cdot, \cdot])$ is called abelian.*

2.4 Examples of Lie Groups, Lie Algebras

In the following two examples, $\mathbb{R}$ can be replaced by $\mathbb{C}$.

Example 2.31. *Some important Lie groups*

1. $GL(n, \mathbb{R}) = \{g \in M_n(\mathbb{R}) : \det g \neq 0\}$
2. $SL(n, \mathbb{R}) = \{g \in GL(n, \mathbb{R}) : \det g = 1\}$
3. $O(n, \mathbb{R}) = \{g \in GL(n, \mathbb{R}) : {}^t g g = I_n\}$
4. $SO(n, \mathbb{R}) = O(n, \mathbb{R}) \cap SL(n, \mathbb{R})$
5. $U(n) = \{g \in GL(n, \mathbb{C}) : {}^t \bar{g} g = I_n\}$
6. $SU(n) = U(n) \cap SL(n, \mathbb{C})$

Example 2.32. *Some important Lie algebras*

1. $\mathfrak{gl}(V)$ *and* $\mathfrak{gl}(n, K)$*, respectively corresponding to* $End_{\mathbb{K}}(V)$ *and* $M_n(K)$*, where K is a field and V is a vector space over K*
2. $\mathfrak{sl}(n, \mathbb{R}) = \{X \in \mathfrak{gl}(n, \mathbb{R}) : tr(X) = 0\}$

3. $\mathfrak{so}(n,\mathbb{R}) = \{X \in \mathfrak{gl}(n,\mathbb{R}) : {}^tX + X = 0\}$
4. $\mathfrak{u}(n,\mathbb{R}) = \{X \in \mathfrak{gl}(n,\mathbb{C}) : {}^t\bar{X} + X = 0\}$
5. $\mathfrak{su}(n) = \mathfrak{u}(n) \cap \mathfrak{sl}(n,\mathbb{C})$

Todo: need a more complete list, with classical Lie groups and algebras.

In particular for this project, we are focusing on the Lie groups $SL(2)$, $SU(2)$, H, $SE(2)$, and $SH(2)$.

1. $SL(2,\mathbb{R}) = \{X \in M_2(\mathbb{R}) : \det A = 1\}$.
2. $SU(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} : \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1 \right\}$.
3. $H = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$ is the Heisenberg group. More generally, we can write $H(A)$, where A is a unitary ring.
4. $SE(2)$ includes arbitrary combinations of translations and rotations in Euclidean space, but not reflections (to be distinguished with $E(n)$). These transformations preserve Euclidean distance.
5. $SH(2)$ is the same as $SE(2)$, except "rotation" is replaced by "hyperbolic rotation". These transformations preserve the Lorentz form.

2.5 Killing Form

A property of finite dimensional Lie algebras is that they come with a symmetric bilinear form, which will be important in the structure theory of Lie algebras.

Definition 2.33. *Let $\mathfrak{g}$ be a finite dimensional Lie algebra. The Killing form of $\mathfrak{g}$ is the bilinear form B on $\mathfrak{g}$ defined by*

$$B(X,Y) = tr(ad(X) \circ ad(Y)), \quad \forall X, Y \in \mathfrak{g}$$

Proposition 2.34. *Let $\mathfrak{g}$ be a finite dimensional Lie algebra, and B its Killing form. Then B is symmetric and ad-invariant, i.e.*

$$B(ad(X)Y, Z) + B(Y, ad(X)Z) = 0, \quad \forall X, Y, Z \in \mathfrak{g}$$

Perhaps a clearer way to express the symmetry property is to write $B(X,Y) = B(Y,X)$. Moreover, ad-invariance means $ad(\mathfrak{g})$ is included in the Lie algebra $\mathfrak{o}(B)$ of the Lie group $O(B)$, which is the group of linear isomorphisms of $\mathfrak{g}$ preserving B.

Proposition 2.35. *Let G be a Lie group with Lie algebra $\mathfrak{g}$. For all $g \in G$, the adjoint map $Ad(g)$ preserves the Killing form of $\mathfrak{g}$, i.e.*

$$B(Ad(g)X, Ad(g)Y) = B(X,Y), \quad \forall X, Y \in \mathfrak{g}$$

The proof of this proposition makes use of the property that

$$ad(Ad(g)X) = Ad(g) \circ ad(x) \circ Ad(g)^{-1}$$

as well as the property that the trace is invariant by conjugation in $GL(\mathfrak{g})$.

It is useful to note that the adjoint map $ad(X) : \mathfrak{g} \to \mathfrak{g}$ is given by

$$ad(X)Y = [X,Y]$$

I also include the definition of the centre of a Lie group or Lie algebra.

Definition 2.36. *The centre of G is*

$$Z(G) = \{g \in G : \forall x \in G \; gx = xg\}$$

Definition 2.37. *The centre of $\mathfrak{g}$ is*

$$\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} : \forall Y \in \mathfrak{g} \; [X, Y] = 0\}$$

Example 2.38. *Some examples of Killing forms:*

1. *The Killing form of $\mathfrak{gl}(n, \mathbb{R})$ is $B(X, Y) = 2ntr(XY) - 2tr(X)tr(Y)$.*
2. *The Killing form of $\mathfrak{sl}(n, \mathbb{R})$ is $B(X, Y) = 2ntr(XY)$.*
3. *The Killing form of an Abelian Lie algebra vanishes.*

Todo: add computational examples of finding the Killing form and the matrix of the Killing form.

2.6 Control Systems

Given two control systems, we can use the following theorem to characterize their equivalence.

Theorem 2.39. *Affine control systems are equivalent if and only if their associated affine distributions are diffeomorphic.*

Let G be a Lie group, and let $\mathfrak{g}$ be its associated Lie algebra. Consider the control system 1.1 with $g \in G$ and $A, B \in \mathfrak{g}$. When classifying 1.1 on this particular Lie group, we want to simplify the pair (A, B) into an equivalent pair, say (A', B'). This equivalency is denoted $(A, B) \sim (A', B')$. What tools can we use in this classification?

The first is the adjoint action of a Lie group on its Lie algebra, $ad : G \times \mathfrak{g} \to \mathfrak{g}$. For each $g \in G$, it is the derivative of this action in the second argument at the neutral element of G. For a matrix Lie group G, it is simply the conjugation $ad_g X = gXg^{-1}$. After letting g act on both A and B, the Killing form $B(A, B)$ is invariant

$$B(gAg^{-1}, gBg^{-1}) = B(A, B)$$

Thus we can write the equivalency $(A, B) \sim (gAg^{-1}, gBg^{-1})$. In different Lie groups, the conjugation group action will have different interpretations, which will become clear in the following sections.

Second, we can rewrite the control u as $u = u' + \delta$ for any δ. This can be thought of as a perturbation to the control. Then,

$$A + uB = A + (u' + \delta)B = (A + \delta B) + u'B$$

and so we have the equivalence $(A, B) \sim ((A + \delta B), B)$.

Another way to modify the control is to scale the control by a constant, writing $u = \mu u'$. Then,

$$A + uB = A + (\mu u')B = A + u'(\mu B)$$

which shows the equivalence $(A, B) \sim (A, \mu B)$.

Finally, the fourth option is to perform time scaling, for example by choosing a new measurement of time. If we scale the time by a factor of $1/\lambda$, then there is the equivalence $(A, B) \sim (\lambda A, \lambda B)$. Clearly, $\lambda \neq 0$.

3 Classifying Control Systems

3.1 Classifying Control Systems on $SU(2)$

We consider the system 1.1 with $g \in SU(2)$ and $A, B \in \mathfrak{su}(2)$. This is the most basic scenario because it is closely related to the usual Euclidean space $\mathbb{R}^3$.

The following matrices

$$u_1 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}; \quad u_2 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}; \quad u_3 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

form a basis for $\mathfrak{su}(2)$. The structural constants are given by the following relations

$$[u_1, u_2] = u_3; \quad [u_2, u_3] = u_1; \quad [u_3, u_1] = u_2$$

The factor of 1/2 is useful in that the Killing form $B(u_i, u_j) = \delta_{ij}$ instead of $4\delta_{ij}$, where δ_{ij} is the Kronecker delta function.

3.2 Classifying Control Systems on $SL(2)$

We consider the system 1.1 with $g \in SL(2, \mathbb{R})$ and $A, B \in \mathfrak{sl}(2)$. The following matrices

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}; \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

form a basis for $\mathfrak{sl}(2)$. The structural constants are given by the following relations

$$[H, E] = 2E; \quad [H, F] = -2F; \quad [E, F] = H$$

This is the most commonly seen choice of basis, and it is the one I will use, for the most part. Note, however, that we could have alternatively chosen the basis

$$E = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}; \quad F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad H = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

which can make the geometry of transformations more apparent.

3.3 Classifying Control Systems on H

We consider the system 1.1 with $g \in H^3$ and $A, B \in \mathfrak{h}$. The following matrices

$$X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}; \quad Z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

form a basis for $\mathfrak{h}$. The structural constants are given by the following relations

$$[X, Y] = Z; \quad [X, Z] = 0; \quad [Y, Z] = 0$$

The Heisenberg group is different from the previous two scenarios in that for any $B \in \mathfrak{h}$, it can be verified that the Killing form $B(B, B) = 0$. So the Killing form is not so useful for classification here.

We start by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z$$

$$B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

There are two types of transformations which will be useful. The first are the inner automorphisms. For $M \in \mathfrak{h}$ and $g \in H$, inner automorphisms are mappings of the form $M \mapsto gMg^{-1}$. If we write M as

$$M = m_1 X + m_2 Y + m_3 Z = \begin{pmatrix} 0 & m_1 & m_3 \\ 0 & 0 & m_2 \\ 0 & 0 & 0 \end{pmatrix}$$

and g as

$$g = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

then the inner automorphism can be computed as

$$\begin{aligned} gMg^{-1} &= \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & m_1 & m_3 \\ 0 & 0 & m_2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -a & ab - c \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & m_1 & -bm_1 + am_2 + m_3 \\ 0 & 0 & m_2 \\ 0 & 0 & 0 \end{pmatrix} \\ &= m_1 X + m_2 Y + (-bm_1 + am_2 + m_3) Z \end{aligned}$$

Notice that the inner automorphism preserves the X and Y components and only shifts the Z component. Applying this transformation to A and B, we obtain

$$gAg^{-1} = a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0) Z$$

$$gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0) Z$$

The second type of transformations are the symplectic maps. The symplectic map uses a matrix $T \in Sp(2, \mathbb{R})$ to act by left matrix multiplication on the (X, Y) vector in M, and preserves the Z component. In symbols, we can write the mapping

$$((X, Y), Z) \mapsto (T(X, Y), Z)$$

Note that in the $n = 2$ case, $Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$. So when $T \in Sp(2, \mathbb{R})$, we can take T to be the rotation matrix

$$T = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Doing so allows us to rotate in the (X, Y) plane.

The result of applying the inner automorphism to A and B - X and Y preserved with Z shifted - suggests that we should consider the centre of the Lie algebra $\mathfrak{h}$. By applying the definition of the centre of a Lie algebra, I found

$$\mathfrak{z}(\mathfrak{h}) = \left\{ \begin{pmatrix} 0 & 0 & c \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : c \in \mathbb{R} \right\} = \{cZ : c \in \mathbb{R}\}$$

We try to simplify B first, and to do so, we analyze two cases: either B belongs to the centre, or B does not belong to the centre.

Case 1: $B \in \mathfrak{z}(\mathfrak{h})$

Here, $b_3^0 \neq 0$, while $b_1^0 X + b_2^0 Y = 0$, i.e. $b_1^0 = b_2^0 = 0$. Then,

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_3^0 Z) \\ &\sim (a_1^0 X + a_2^0 Y, b_3^0 Z) \end{aligned}$$

If $(a_1^0, a_2^0) = (0, 0)$, then

$$\begin{aligned} (A, B) &\sim (0, b_3^0 Z) \\ &\sim (0, Z) \end{aligned}$$

If $(a_1^0, a_2^0) \neq (0, 0)$, then using the rotation matrix T simultaneously on A and B, we can rotate $a_1^0 X + a_2^0 Y$ on to either X or Y, while $b_3^0 Z$ remains unchanged. I choose to rotate on to X, and so

$$\begin{aligned} (A, B) &\sim (a_1^1 X, b_3^0 Z) \\ &\sim (a_1^1 X, a_1^1 Z) \\ &\sim (X, Z) \end{aligned}$$

Case 2: $B \notin \mathfrak{z}(\mathfrak{h})$

Here, $b_1^0 X + b_2^0 Y \neq 0$, so $(b_1^0, b_2^0) \neq (0, 0)$. b_3^0 may or may not be 0. In the inner automorphism on B,

$$gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0)Z$$

since $(b_1^0, b_2^0) \neq (0, 0)$, we can choose the entries $a, b \in \mathbb{R}$ of the matrix g such that $-bb_1^0 + ab_2^0 + b_3^0 = 0$, and so

$$gBg^{-1} = b_1^0 X + b_2^0 Y$$

This can further be rotated on to X by applying the symplectic map to A and B. We have the following

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \end{aligned}$$

If $a_2^1 = 0$, then

$$\begin{aligned}(A,B) &= ((-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X)\\ &= ((-ba_1^0 + aa_2^0 + a_3^0)Z, (-ba_1^0 + aa_2^0 + a_3^0)X)\\ &= (Z, X)\end{aligned}$$

If $a_2^1 \neq 0$, then we need another inner automorphism that is at the same time an element of the stabiliser of X. Take another $h \in H$ of the form

$$h = \begin{pmatrix} 1 & a' & c' \\ 0 & 1 & b' \\ 0 & 0 & 1 \end{pmatrix}$$

For h to be an element of $Stab(X)$, we require $hXh^{-1} = X$. Computing this gives

$$\begin{pmatrix} 0 & 1 & -b' \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

thus we need $b' = 0$ to ensure $hXh^{-1} = X$. Moreover,

$$hYh^{-1} = \begin{pmatrix} 0 & 0 & a' \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = Y + a'Z$$

$$hZh^{-1} = Z$$

We now apply this inner automorphism.

$$\begin{aligned}(A,B) &\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X)\\ &\sim (h(a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z)h^{-1}, h(b_1^1 X)h^{-1})\\ &= (a_2^1(Y + a'Z) + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X)\\ &= (a_2^1 Y + [a_2^1 a' + (-ba_1^0 + aa_2^0 + a_3^0)]Z, b_1^1 X)\end{aligned}$$

Since $a_2^1 \neq 0$, we can choose the entries of h to be $a' = -(-ba_1^0 + aa_2^0 + a_3^0)/a_2^1$, $b' = 0$, $c' \in \mathbb{R}$ so that $a_2^1 a' + (-ba_1^0 + aa_2^0 + a_3^0) = 0$. Proceeding with this,

$$\begin{aligned}(A,B) &\sim (a_2^1 Y, b_1^1 X)\\ &\sim (a_2^1 Y, a_2^1 X)\\ &\sim (Y, X)\end{aligned}$$

3.4 Classifying Control Systems on $SE(2)$

The $SE(2)$ group is defined as follows

$$SE(2) = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{pmatrix} : \theta, x, y \in \mathbb{R} \right\}$$

I will derive the corresponding Lie algebra here, demonstrating a general method for doing so. To obtain the identity element of $SE(2)$, we need $\theta = 0$, $x = 0$, and $y = 0$. Parameterize $\theta(t)$, $x(t)$, $y(t)$ with $t \geq 0$ such that $\theta(0) = 0$, $x(0) = 0$, $y(0) = 0$. That is, these curves pass through the identity at $t = 0$. Then, we differentiate an element of $SE(2)$ at $t = 0$.

$$\begin{aligned}
\frac{d}{dt} \begin{pmatrix} \cos\theta(t) & -\sin\theta(t) & x(t) \\ \sin\theta(t) & \cos\theta(t) & y(t) \\ 0 & 0 & 1 \end{pmatrix} \Bigg|_{t=0} &= \begin{pmatrix} -\sin\theta(t)\dot{\theta}(t) & -\cos\theta(t)\dot{\theta}(t) & \dot{x}(t) \\ \cos\theta(t)\dot{\theta}(t) & -\sin\theta(t)\dot{\theta}(t) & \dot{y}(t) \\ 0 & 0 & 0 \end{pmatrix} \Bigg|_{t=0} \\
&= \begin{pmatrix} 0 & -\dot{\theta}(0) & \dot{x}(0) \\ \dot{\theta}(0) & 0 & \dot{y}(0) \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & -c & a \\ c & 0 & a \\ 0 & 0 & 0 \end{pmatrix}
\end{aligned}$$

Thus, the Lie algebra corresponding to $SE(2)$ is

$$\mathfrak{se}(2) = \left\{ \begin{pmatrix} 0 & -c & a \\ c & 0 & a \\ 0 & 0 & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$$

The following matrices

$$X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}; \quad Z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

form a basis for $\mathfrak{se}(2)$.

Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z$$

$$B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SE(2)$, we have

$$g = \begin{pmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{pmatrix}, \quad g^{-1} = \begin{pmatrix} \cos\theta & \sin\theta & -x\cos\theta - y\sin\theta \\ -\sin\theta & \cos\theta & x\sin\theta - y\cos\theta \\ 0 & 0 & 1 \end{pmatrix}$$

where $\theta, x, y \in \mathbb{R}$. The conjugation group action on the basis elements gives

$$gXg^{-1} = \begin{pmatrix} 0 & 0 & \cos\theta \\ 0 & 0 & \sin\theta \\ 0 & 0 & 0 \end{pmatrix} = \cos\theta X + \sin\theta Y$$

$$gYg^{-1} = \begin{pmatrix} 0 & 0 & -\sin\theta \\ 0 & 0 & \cos\theta \\ 0 & 0 & 0 \end{pmatrix} = -\sin\theta X + \cos\theta Y$$

$$gZg^{-1} = \begin{pmatrix} 0 & -1 & y \\ 1 & 0 & -x \\ 0 & 0 & 0 \end{pmatrix} = yX - xY + Z$$

Either making use of the decomposition into basis elements or by direct calculation, we have

$$gAg^{-1} = (a_1^0 \cos\theta - a_2^0 \sin\theta + a_3^0 y)X + (a_1^0 \sin\theta + a_2^0 \cos\theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cos\theta - b_2^0 \sin\theta + b_3^0 y)X + (b_1^0 \sin\theta + b_2^0 \cos\theta - b_3^0 x)Y + b_3^0 Z$$

Indeed, the transformations here are combinations of rotations and translations in Euclidean space, where rotations are determined by θ, and translations are determined by x and y.

There are two cases to consider: when $B \in Span(X, Y)$, and when $B \notin Span(X, Y)$.

Case 1: $B \in Span(X, Y)$

In this case, $B = b_1^0 X + b_2^0 Y$, with $(b_1^0, b_2^0) \neq 0$ and $b_3^0 = 0$. Since b_3^0, we can disregard x and y, and choose θ so that gBg^{-1} rotates B in the (X, Y) plane on to X, then simplify.

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= ((a_1^0 \cos\theta - a_2^0 \sin\theta + a_3^0 y)X + (a_1^0 \sin\theta + a_2^0 \cos\theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X) \\
&= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X) \\
&\sim (a_2^1 Y + a_3^0 Z, b_1^1 X)
\end{aligned}$$

If $a_3^0 = 0$, then assuming $a_2^1 \neq 0$,

$$\begin{aligned}
(A, B) &\sim (a_2^1 Y, b_1^1 X) \\
&\sim (a_2^1 Y, a_2^1 X) \\
&\sim (Y, X)
\end{aligned}$$

If $a_3^0 \neq 0$, then we need to find the elements of the stabiliser group of X, so that applying gAg^{-1} and gBg^{-1} preserves $b_1^1 X$. The stabiliser is

$$Stab(X) = \{g \in SE(2) : gXg^{-1} = X\}$$

From the result of gXg^{-1} computed above, it is clear that to preserve $b_1^1 X$, we just need $\theta = 0$. Moreover, noting the result of gZg^{-1}, we choose $x = a_3^1/a_3^0$ and $y = 0$ to eliminate the Y component and prevent X from reappearing in A. Thus,

$$\begin{aligned}
(A, B) &\sim (a_3^0 Z, b_1^1 X) \\
&\sim (a_3^0 Z, a_3^0 X) \\
&\sim (Z, X)
\end{aligned}$$

Case 2: $B \notin Span(X, Y)$

Here, $b_3^0 \neq 0$. When applying gAg^{-1} and gBg^{-1}, we can choose $x = 0$ and $y = 0$ so that we are just rotating in the (X, Y) plane.

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= ((a_1^0 \cos\theta - a_2^0 \sin\theta + a_3^0 y)X + (a_1^0 \sin\theta + a_2^0 \cos\theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X + b_3^0 Z) \\
&= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X + b_3^0 Z)
\end{aligned}$$

If $b_1^1 = 0$, i.e. $B \in Span(Z)$, then

$$\begin{aligned}
(A, B) &\sim (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\
&\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)
\end{aligned}$$

Take another $g \in SE(2)$ and choose $x = 0$, $y = 0$, and $\theta \in \mathbb{R}$ to perform a rotation on to X, while Z is preserved.

$$\begin{aligned}(A, B) &\sim (a_1^2 X, b_3^0 Z)\\ &\sim (a_1^2 X, a_1^2 Z)\\ &\sim (X, Z)\end{aligned}$$

If $b_1^1 \neq 0$, we take another $g \in SE(2)$ with $\theta = 0$ so that there is no rotation, only translation. We compute

$$g(a_1^1 X + a_2^1 Y + a_3^0 Z)g^{-1} = (a_1^1 + a_3^0 y)X + (a_2^1 - a_3^0 x)Y + a_3^0 Z$$

$$g(b_1^1 X + b_3^0 Z)g^{-1} = (b_1^1 + b_3^0 y)X - b_3^0 xY + b_3^0 Z$$

Choosing $x = 0$, $y = -b_1^1/b_3^0$ gives

$$\begin{aligned}(A, B) &\sim (a_1^2 X + a_2^1 Y + a_3^0 Z, b_3^0 Z)\\ &\sim (a_1^2 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

Finally, take another $g \in SE(2)$ and set $x = 0$, $y = 0$. $\theta \in \mathbb{R}$ can be chosen such that $a_1^2 X + a_2^1 Y$ is rotated on to X.

$$\begin{aligned}(A, B) &\sim (g(a_1^2 X + a_2^1 Y)g^{-1}, g(b_3^0 Z)g^{-1})\\ &= (a_1^3 X, b_3^0 Z)\\ &\sim (a_1^3 X, a_1^3 Z)\\ &\sim (X, Z)\end{aligned}$$

3.5 Classifying Control Systems on $SH(2)$

The $SH(2)$ group is similar to $SE(2)$. While $SE(2)$ is translations and rotations in Euclidean space preserving Euclidean distance, $SH(2)$ is translations and hyperbolic rotations preserving the Lorentz form. It is given as follows

$$SH(2) = \left\{ \begin{pmatrix} \cosh\theta & \sinh\theta & x \\ \sinh\theta & \cosh\theta & y \\ 0 & 0 & 1 \end{pmatrix} : \theta, x, y \in \mathbb{R} \right\}$$

Recall that sinh and cosh are the hyperbolic trigonometric functions defined by

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

with the relation

$$\cosh^2 x - \sinh^2 x = 1$$

In the following, it will be useful to remember that $\sinh 0 = 0$ and $\cosh 0 = 1$.

In a similar fashion to the $SE(2)$ case, the Lie algebra corresponding to the $SH(2)$ group can be derived to be

$$\mathfrak{sh}(2) = \left\{ \begin{pmatrix} 0 & c & a \\ c & 0 & a \\ 0 & 0 & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$$

The matrices

$$X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}; \quad Z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

form a basis for $\mathfrak{sh}(2)$.

Again, begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z$$

$$B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SH(2)$, we have

$$g = \begin{pmatrix} \cosh\theta & \sinh\theta & x \\ \sinh\theta & \cosh\theta & y \\ 0 & 0 & 1 \end{pmatrix}, \quad g^{-1} = \begin{pmatrix} \cosh\theta & -\sinh\theta & -x\cosh\theta + y\sinh\theta \\ -\sinh\theta & \cosh\theta & x\sinh\theta - y\cosh\theta \\ 0 & 0 & 1 \end{pmatrix}$$

where $\theta, x, y \in \mathbb{R}$. The conjugation group action on the basis elements gives

$$gXg^{-1} = \begin{pmatrix} 0 & 0 & \cosh\theta \\ 0 & 0 & \sinh\theta \\ 0 & 0 & 0 \end{pmatrix} = \cosh\theta X + \sinh\theta Y$$

$$gYg^{-1} = \begin{pmatrix} 0 & 0 & \sinh\theta \\ 0 & 0 & \cosh\theta \\ 0 & 0 & 0 \end{pmatrix} = \sinh\theta X + \cosh\theta Y$$

$$gZg^{-1} = \begin{pmatrix} 0 & 1 & -y \\ 1 & 0 & -x \\ 0 & 0 & 0 \end{pmatrix} = -yX - xY + Z$$

Either making use of the decomposition into basis elements or by direct calculation, we have

$$gAg^{-1} = (a_1^0 \cosh\theta + a_2^0 \sinh\theta - a_3^0 y)X + (a_1^0 \sinh\theta + a_2^0 \cosh\theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cosh\theta + b_2^0 \sinh\theta - b_3^0 y)X + (b_1^0 \sinh\theta + b_2^0 \cosh\theta - b_3^0 x)Y + b_3^0 Z$$

Indeed, the transformations here are combinations of hyperbolic rotations determined by θ, and translations determined by x and y.

The analysis requires two cases to be considered: when $B \in Span(X, Y)$ and when $B \notin Span(X, Y)$.

Case 1: $B \in Span(X, Y)$

Here, $b_3^0 = 0$ and $(b_1^0, b_2^0) \neq (0, 0)$, so we have

$$gAg^{-1} = (a_1^0 \cosh\theta + a_2^0 \sinh\theta - a_3^0 y)X + (a_1^0 \sinh\theta + a_2^0 \cosh\theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cosh\theta + b_2^0 \sinh\theta)X + (b_1^0 \sinh\theta + b_2^0 \cosh\theta)Y$$

If $b_1^0 \neq 0$ and $b_2^0 = 0$, then

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_1^0 X)\\ &\sim (a_2^0 Y + a_3^0 Z, b_1^0 X)\end{aligned}$$

Now, if $a_3^0 = 0$, then

$$\begin{aligned}(A, B) &\sim (a_2^0 Y, b_1^0 X)\\ &\sim (a_2^0 Y, a_2^0 X)\\ &\sim (Y, X)\end{aligned}$$

If $a_3^0 \neq 0$, then we apply a transformation consisting only of a shift in Y. This is done by choosing $\theta = 0$ so there is no rotation, $y = 0$ so there is no shift in X, and then taking $x = a_2^0/a_3^0$ to shift Y to zero. This is demonstrated as follows

$$\begin{aligned}(A, B) &\sim (g(a_2^0 Y + a_3^0 Z)g^{-1}, g(b_1^0 X)g^{-1})\\ &= ((a_2^0 - a_3^0 x)Y + a_3^0 Z, b_1^0 X)\\ &= (a_3^0 Z, b_1^0 X)\\ &\sim (a_3^0 Z, a_3^0 X)\\ &\sim (Z, X)\end{aligned}$$

If $b_1^0 = 0$ and $b_2^0 \neq 0$, then

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_2^0 Y)\\ &\sim (a_1^0 X + a_3^0 Z, b_2^0 Y)\end{aligned}$$

If $a_3^0 = 0$, then

$$\begin{aligned}(A, B) &\sim (a_1^0 X, b_2^0 Y)\\ &\sim (a_1^0 X, a_1^0 Y)\\ &\sim (X, Y)\end{aligned}$$

If $a_3^0 \neq 0$, then we apply a transformation consisting only of a shift in X. This is done by choosing $\theta = 0$ so there is no rotation, $x = 0$ so there is no shift in Y, and then taking $y = a_1^0/a_3^0$ to shift X to zero.

$$\begin{aligned}(A, B) &\sim (g(a_1^0 X + a_3^0 Z)g^{-1}, g(b_2^0 Y)g^{-1})\\ &= ((a_1^0 - a_3^0 y)X + a_3^0 Z, b_2^0 Y)\\ &= (a_3^0 Z, b_2^0 Y)\\ &\sim (a_3^0 Z, a_3^0 Y)\\ &\sim (Z, Y)\end{aligned}$$

The most general case is $b_1^0 \neq 0$ and $b_2^0 \neq 0$. The main idea here is that we will try to use hyperbolic rotations to simplify B. Referring to the form of B after a hyperbolic rotation, given by

$$gBg^{-1} = (b_1^0 \cosh\theta + b_2^0 \sinh\theta)X + (b_1^0 \sinh\theta + b_2^0 \cosh\theta)Y,$$

we see that if we want to reduce one of the coefficients to 0, we need to solve for θ in $\tanh\theta = -b_1^0/b_2^0$ or $-b_2^0/b_1^0$. tanh is a mapping $\mathbb{R} \mapsto (-1,1)$. There is no solution when $b_1^0 = b_2^0$. So three different cases need to be analyzed, depending on the ratio of b_1^0 and b_2^0.

If $b_1^0 = b_2^0$, then

$$\begin{aligned}(A,B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1X + a_2^1Y + a_3^0Z, b_1^0X + b_1^0Y)\end{aligned}$$

In this case, if $a_3^0 = 0$, then

$$\begin{aligned}(A,B) &\sim (a_1^1X + a_2^1Y, b_1^0X + b_1^0Y) \\ &\sim (a_1^2X, b_1^0(X+Y)) \\ &\sim (a_1^2X, a_1^2(X+Y)) \\ &\sim (X, X+Y)\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, then take another $g \in SH(2)$ with $\theta = 0$ so that there is no rotation. Applying the transformation gives

$$\begin{aligned}(A,B) &\sim (g(a_1^1X + a_2^1Y + a_3^0Z)g^{-1}, g(b_1^0X + b_1^0Y)g^{-1}) \\ &\sim ((a_1^1 - a_3^0y)X + (a_2^1 - a_3^0x)Y + a_3^0Z, b_1^0(X+Y))\end{aligned}$$

It is possible to choose $x, y \in \mathbb{R}$ such that $a_1^1 - a_3^0y = 0$ and $a_2^1 - a_3^0x = 0$. Hence

$$\begin{aligned}(A,B) &\sim (a_3^0Z, b_1^0(X+Y)) \\ &\sim (a_3^0Z, a_3^0(X+Y)) \\ &\sim (Z, X+Y)\end{aligned}$$

If $b_1^0 < b_2^0$, then the equation $\tanh\theta = -b_1^0/b_2^0$ has a solution. There exists a $\theta \in \mathbb{R}$ such that $b_1^0 \cosh\theta + b_2^0 \sinh\theta = 0$, so $gBg^{-1} = b_2^1Y$.

$$\begin{aligned}(A,B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1X + a_2^1Y + a_3^0Z, b_2^1Y) \\ &\sim (a_1^1X + a_3^0Z, b_2^1Y)\end{aligned}$$

In this case, if $a_3^0 = 0$, then

$$\begin{aligned}(A,B) &\sim (a_1^1X, b_2^1Y) \\ &\sim (a_1^1X, a_1^1Y) \\ &\sim (X, Y)\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, use another $g \in SH(2)$

$$\begin{aligned}(A,B) &\sim (g(a_1^1X + a_3^0Z)g^{-1}, g(b_2^1Y)g^{-1}) \\ &= ((a_1^1 - a_3^0y)X + a_3^0Z, b_2^1Y) \\ &= (a_3^0Z, b_2^1Y) \\ &\sim (a_3^0Z, a_3^0Y) \\ &\sim (Z, Y)\end{aligned}$$

where I chose $\theta = 0$ so that there is no rotation, $x = 0$ so that Y is unchanged, and $y = a_1^1/a_3^0$ so that the coefficient $a_1^1 - a_3^0 y$ vanishes. This is the same idea as seen before. The two equivalent forms here have already been encountered.

If $b_2^0 < b_1^0$, then the equation $\tanh\theta = -b_2^0/b_1^0$ has a solution. There exists a $\theta \in \mathbb{R}$ such that $b_1^0 \sinh\theta + b_2^0 \cosh\theta = 0$, so $gBg^{-1} = b_1^1 X$.

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X) \\ &\sim (a_2^1 Y + a_3^0 Z, b_1^1 X)\end{aligned}$$

In this case, if $a_3^0 = 0$, then

$$\begin{aligned}(A, B) &\sim (a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, a_2^1 X) \\ &\sim (Y, X)\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, use another $g \in SH(2)$

$$\begin{aligned}(A, B) &\sim (g(a_2^1 Y + a_3^0 Z)g^{-1}, g(b_1^1 X)g^{-1}) \\ &= ((a_2^1 - a_3^0 x)Y + a_3^0 Z, b_1^1 X) \\ &= (a_3^0 Z, b_1^1 X) \\ &\sim (a_3^0 Z, a_3^0 X) \\ &\sim (Z, X)\end{aligned}$$

where I chose $\theta = 0$ so that there is no rotation, $y = 0$ so that X is unchanged, and $x = a_2^1/a_3^0$ so that the coefficient $a_2^1 - a_3^0 x$ vanishes. Again, this is the same idea as seen before. The two equivalent forms here have already been encountered as well.

Case 2: $B \notin Span(X, Y)$

Here, $b_3^0 \neq 0$. Recalling the general form of gAg^{-1} and gBg^{-1} that was initially derived, it is possible to choose $x, y \in \mathbb{R}$ such that the coefficients $b_1^0 \cosh\theta + b_2^0 \sinh\theta - b_3^0 y = 0$ and $b_1^0 \sinh\theta + b_2^0 \cosh\theta - b_3^0 x = 0$, causing the X and Y components in B to vanish. We have

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 = 0$, then

$$\begin{aligned}(A, B) &\sim (a_1^1 X, b_3^0 Z) \\ &\sim (a_1^1 X, a_1^1 Z) \\ &\sim (X, Z)\end{aligned}$$

If $a_1^1 = 0$ and $a_2^1 \neq 0$, then

$$\begin{aligned}(A, B) &\sim (a_2^1 Y, b_3^0 Z) \\ &\sim (a_2^1 Y, a_2^1 Z) \\ &\sim (Y, Z)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 \neq 0$, then just as before, there are three cases to analyze. If $a_1^1 = a_2^1$, then

$$\begin{aligned}(A, B) &\sim (a_1^1(X + Y), b_3^0 Z) \\ &\sim (a_1^1(X + Y), a_1^1 Z) \\ &\sim (X + Y, Z)\end{aligned}$$

Otherwise, in order to simplify further, we need to choose a $g \in SH(2)$ that is also an element of the stabiliser group $Stab(Z)$, where

$$Stab(Z) = \{g \in SH(2) : gZg^{-1} = Z\}$$

It is clear that we require $x = 0$ and $y = 0$. We have

$$\begin{aligned}(A, B) &\sim (g(a_1^1 X + a_2^1 Y)g^{-1}, g(b_3^0 Z)g^{-1}) \\ &= ((a_1^1 \cosh\theta + a_2^1 \sinh\theta)X + (a_1^1 \sinh\theta + a_2^1 \cosh\theta)Y, b_3^0 Z)\end{aligned}$$

We follow the same reasoning as before and try to apply rotations to simplify. If $a_1^1 < a_2^1$, then there exists a $\theta \in \mathbb{R}$ such that the coefficient $a_1^1 \cosh\theta + a_2^1 \sinh\theta = 0$. Hence

$$\begin{aligned}(A, B) &\sim (a_2^2 Y, b_3^0 Z) \\ &\sim (a_2^2 Y, a_2^2 Z) \\ &\sim (Y, Z)\end{aligned}$$

On the other hand, if $a_2^1 < a_1^1$, then there exists a $\theta \in \mathbb{R}$ such that the coefficient $a_1^1 \sinh\theta + a_2^1 \cosh\theta = 0$. Hence

$$\begin{aligned}(A, B) &\sim (a_1^2 X, b_3^0 Z) \\ &\sim (a_1^2 X, a_1^2 Z) \\ &\sim (X, Z)\end{aligned}$$

3.6 Summary

The following table summarizes the model control systems to which control systems over each Lie group are equivalent.

$SU(2)$	(u_1, u_3)
$SL(2)$	(E, H)
	(F, H)
	$(\pm E \pm F, H)$
	$(E, E - F)$
	(H, E)
	(F, E)
H	(X, Z)
	(Z, X)
	(Y, X)
$SE(2)$	(Y, X)
	(Z, X)
	(X, Z)
$SH(2)$	(X, Z)
	(Y, Z)
	$(X + Y, Z)$
	$(X, X + Y)$
	$(Z, X + Y)$
	(Z, Y)
	(X, Y)
	(Z, X)
	(Y, X)

Table 1: Summary of Model Control Systems